

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1539

Renaissance 100 kV
October 28, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1539. The Interconnection Customer proposes a 216.0 MW solar photovoltaic facility interconnecting at the Renaissance 100 kV substation in the Illinois area. The request was received on August 03, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 536 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$75,332,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FIELDER CREEK – APISON PIKE 100kV line	HILL COUNTRY 138 230/161kV Transformer Bank	186.5	130	64.8
A EAST TERM – JACKSON-SUNRISE 100kV line	HILL COUNTRY 138 230 kV/161 kV XFMR	185.3	146	64.8
APISON PIKE – A E TERM 100kV line	HILL COUNTRY 138 230/161kV Autotransformer	184.2	140	64.8
DRAKE SUBSTATION – GEORGE RD 100kV	HILL COUNTRY 138 230-161 kV Transformer Bank	167.3	160	77.1
HILL COUNTRY 138 230-161 kV Autotransformer	A E TERM – JACKSON-SUNRISE 100kV	142.2	195	73.5
KEAHOE – CARGILL WIRE 100kV	HILL COUNTRY 138 230-161 kV TX	140.2	100	35.2
STERLING STEEL SUBSTATION – KEAHOE 100kV	HILL COUNTRY 138 230/161 kV TX	133.1	121	35.2
JACKSON-SUNRISE – GEORGE RD 100kV Line	HILL COUNTRY 138 230 kV/161 kV Auto	128.7	184	71.3
HILL COUNTRY 138 161/100 kV Transformer Bank	HILL COUNTRY 38 – QUAD CITIES 161kV	118.6	128	49.2
DRAKE SUBSTATION – RENAISSANCE 100kV Circuit 1	HILL COUNTRY 138 230-161 kV Transformer	108.2	168	50.0
DRAKE – RENAISSANCE SUBSTATION 100kV	HILL COUNTRY 138 230/161kV TX	108.2	168	50.0
BLOUNTVILLE SW STA – STERLING STEEL SUBSTATION 100kV line	JACKSON-SUNRISE – GEORGE RD 100kV	106.4	123	18.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BLOUNTVILLE SW STATION – JACKSON-SUNRISE 100kV L	GEORGE ROAD – JACKSON-SUNRISE 100kV L	102.4	128	18.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FIELDER CREEK – APISON PIKE 100kV	Rebuild line with higher-capacity conductor	\$4,584,000	64% (\$2,927,000)
A E TERM SUBSTATION – JACKSON-SUNRISE 100kV	Rebuild line with higher-capacity conductor	\$2,528,000	64% (\$1,615,000)
APISON PIKE – A E TERM 100kV	Rebuild line with higher-capacity conductor	\$4,167,000	64% (\$2,661,000)
DRAKE SUBSTATION – GEORGE RD 100kV	Rebuild line with higher-capacity conductor	\$4,583,000	100% (\$4,583,000)
HILL COUNTRY 138 230-161 kV TX	Replace transformer with higher-rated unit	\$9,841,000	69% (\$6,822,000)
KEAHOE – CARGILL WIRE 100kV	Rebuild line with higher-capacity conductor	\$34,371,000	59% (\$20,331,000)
STERLING STEEL SUBSTATION – KEAHOE 100kV	Rebuild line with higher-capacity conductor	\$19,948,000	59% (\$11,799,000)
JACKSON-SUNRISE – GEORGE RD 100kV Line	Rebuild line with higher-capacity conductor	\$15,707,000	93% (\$14,552,000)
HILL COUNTRY 138 161/100 kV Transformer Bank	Replace transformer with higher-rated unit	\$5,787,000	76% (\$4,399,000)
DRAKE – RENAISSANCE 100kV Ckt 1	Reconductor line	\$2,551,000	100% (\$2,551,000)
DRAKE – RENAISSANCE SUBSTATION 100kV	Reconductor line	\$2,404,000	100% (\$2,404,000)
STERLING STEEL – BLOUNTVILLE SW STATION 100kV	Reconductor line	\$442,000	40% (\$179,000)
BLOUNTVILLE SW STATION – JACKSON-SUNRISE 100kV L	Reconductor line	\$1,258,000	40% (\$509,000)
	Total allocated to project		\$75,332,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.